Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
Members	Vincentius Bryan Kwandou (leader), Valentino Filemon Limer, Putra Purwanugraha Tengbunan, Marvel Harjosetio

Abstract

1. Problem. Student treasurers file accountability reports (LPJ) before faculty move subsequent proposals. Rules differ by campus: original stamps, one attendance sheet per meal, identical totals, a revenue stamp only when nominal exceeds Rp5,000,000 (Law 10/2020 Art. 3), withholding, and strict deadlines. At FILKOM UB, late or incomplete reports delay proposals. Treasurers lack formal accounting training, spending hours on mechanical rework. Revision rates remain unmeasured; establishing that count is our first validation metric, not an unverified claim.

2. Intended Users. Primary users are treasurers and secretaries of program associations, BEM, and clubs. The faculty desk reviewer reads the output. They are not a second product: RapiLPJ checks a file and does not release money.

3. Supporting Observations or Evidence. FIB UB requires stamped originals, one attendance sheet per meal session, and matching totals. FEB UGM requires a Rp10,000 stamp above Rp5,000,000 and a scan when thermal paper fades. An ITS department sets ten days; ITS Biology wants one line per receipt with tax included. A 2021 UAJM study describes unstandardized activity documentation. FILKOM UB's 2026 circular records repeated withholding and completeness failures.

4. Proposed Solution and Core User Flow. RapiLPJ is a practice run before submission. First, perceive: Gemini Flash on Vertex AI (Singapore) extracts vendor, date, invoice number, nominal, and stamp presence as JSON without arithmetic. Fields sit beside the slip; human changes enter an edit disclosure log. Offline Indonesian fixtures run without API keys. Second, a client-side kernel flags duplicate invoices, out-of-window dates, missing stamps above Rp5,000,000, and variances above 10 percent. Meal caps and PPh 21 use only a rule card signed by two advisors; missing rates return 'preset gap' without inventing figures. Third, export unlocks only when a micro-quiz matches the statutory key—exactly Rp5,000,000 is not dutiable; above is—and a variance addendum is completed via fixed template. Fourth, export yields a dossier with SHA-256 over confirmed ledger, edit log, quiz, addendum, and finding IDs, which verifiers recompute locally.

5. How It Differs from Existing Alternatives. A raw LLM mis-adds. A spreadsheet does not bind a hash, cite the statute, or lock export. Campus portals collect files; they do not pre-audit them.

6. Planned Technologies. Static bilingual HTML, CSS, and JavaScript on Firebase Hosting, no application database. Open JavaScript kernel, FileReader, a .lpj JSON file, and Web Crypto SHA-256. WCAG 2.1 AA.

7. Expected Benefit and Development Plan. The semifinal benchmarks single-activity intake: perception, confirmation, signed card, quiz, and desk-check. Final Day adds multi-slip batching and QR re-checks. With 30 treasurers and 2 advisors we evaluate minutes to a clean dossier, mechanical defects resolved, fields corrected post-perception, and quiz retention without on-screen cues. The primary risk is misread thermal slips; human confirmation plus edit logging provides control. Arithmetic never trusts the model.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119 (studi sistem informasi portofolio kemahasiswaan BKAM UAJM). [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026.